

Nanyang Technological University
IS6751 Text and Web Mining
Lab: Sentiment Analysis with BERT

In this lab, you will learn how to build a sentiment classifier with BERT.

1. Install gdown in Anaconda Prompt (Start | Anaconda3 | Anaconda Prompt)

```
$> pip install gdown
```

2. Running the Jupyter Notebook

2.1. Copy the **notebook** files downloaded from the class website to
C:\Users\youraccountname\documents\python\IS6751

2.2 Open a Jupyter Notebook: Use Google Colab (faster, it will take around 1-2 hours)

- Go to <https://colab.research.google.com/>, and login with your google ID/PW.
- Upload **sentiment_analysis_with_bert_for_colab_v3.ipynb** in
C:\Users\youraccountname\documents\python\IS6751\wk11-sa-bert\
- Make a connection: **Connect to hosted runtime**
- Set the running environment to **GPU: Runtime -> Change runtime type**
- Run the code: **Runtime -> Run all**

2.3 Open a Jupyter Notebook: Use Local Jupyter Notebook (very slow when you use a CPU machine)

- File – Open ..., **sentiment_analysis_with_bert_for_colab_v3.ipynb** in
C:\Users\youraccountname \documents\python\IS6751\wk11-sa-bert\
- Run the notebook document step-by-step (one cell a time)
- **Since training takes some time, stop the training process.**
- **And uncomment and comment code as follows to use the pre-trained model on a CPU machine and re-run the code from the cell.**

```
model.load_state_dict(torch.load('model/best_model_state.bin',  
                                map_location=torch.device('cpu')))  
#model.load_state_dict(torch.load('best_model_state.bin'))  
model = model.to(device)
```